



Introduction to BigQuery ML

Bringing Machine Learning to Your Data

What is BigQuery ML?

In-Database Machine Learning

BigQuery ML (BQML) enables data analysts and scientists to build, train, evaluate, and deploy ML models directly inside BigQuery using standard SQL queries.

Zero Data Movement

Eliminates security risks and overhead of exporting terabytes of data to external training environments.

Speed to Production

Models deploy instantly upon creation. Execute real-time batch or streaming predictions with SQL.

Skill Accessibility

Democratizes machine learning across existing enterprise SQL skill sets beyond specialized ML engineers.



BigQuery ML

SQL for Machine Learning



1. Model Training BigQuery SQL

```
CREATE OR REPLACE MODEL `project.dataset.churn_model`  
OPTIONS(  
  model_type = 'LOGISTIC_REG',  
  input_label_cols = ['has_churned'],  
  auto_class_weights = TRUE  
) AS  
SELECT user_id, total_spend, tenure_days, support_tickets, has_churned  
FROM `project.dataset.user_features`;
```



2. Model Evaluation BigQuery SQL

```
SELECT *  
FROM ML.EVALUATE(  
  MODEL `project.dataset.churn_model`,  
  (SELECT * FROM `project.dataset.test_data`)  
);
```



3. Batch Prediction BigQuery SQL

```
SELECT user_id, predicted_has_churned, predicted_has_churned_probs  
FROM ML.PREDICT(  
  MODEL `project.dataset.churn_model`,  
  (SELECT * FROM `project.dataset.active_users`)  
);
```



4. Generative AI Text Generation BigQuery SQL

```
SELECT *  
FROM ML.GENERATE_TEXT(  
  MODEL `project.dataset.gemini_flash_model`,  
  (SELECT customer_review AS prompt FROM `project.dataset.feedback`),  
  STRUCT(0.2 AS temperature, 256 AS max_output_tokens)  
);
```



Standardized Lifecycle Functions: Native SQL support for MLEVALUATE, MLPREDICT, & MLFEATUREINFO.



Automatic Preprocessing: Automatic categorical encoding, missing value imputation, and scaling.

Modern AI in BigQuery



Remote LLMs & Gemini

Connect BQML directly to Google's Gemini models via Vertex AI endpoints within SQL.

- Text generation & summarization
- Entity extraction & classification
- Sentiment & feedback analysis
- **SQL Function:** MLGENERATE_TEXT



Embeddings Generation

Convert unstructured text, images, or documents into dense vector representations.

- Semantic content analysis
- Unstructured data clustering
- Multimodal vector transformation
- **SQL Function:** MLGENERATE_EMBEDDING



Vector Search in SQL

Perform high-speed vector similarity searches directly inside analytical queries.

- Retrieval-Augmented GenAI (RAG)
- Real-time recommendation engines
- IVF / HNSW indexing for fast k-NN
- **SQL Function:** VECTOR_SEARCH

💡 "BigQuery ML transforms your data warehouse into a unified engine for both structured analytical ML and unstructured Generative AI workflows."

BigQuery ML in GCP Ecosystem



Vertex AI Model Registry

Integration: Auto-register BQML models into Vertex AI with a single SQL flag.

Benefit: Enables end-to-end MLOps, model versioning, and endpoint deployment alongside custom Python models.



Dataform & dbt Integration

Integration: Embed BQML CREATE MODEL and ML.PREDICT in automated SQL pipelines.

Benefit: Manage ML models natively as versioned assets in CI/CD pipelines and transformation DAGs.



Looker & BI Engines

Integration: Query live predictions via standard SQL directly inside BI dashboards.

Benefit: Expose real-time risk scores and recommended actions to business stakeholders without API middleware.



BigQuery Studio & Dataplex

Integration: Embedded Python notebooks (Colab) sharing unified metadata governance.

Benefit: Seamless collaboration and smooth hand-off between SQL data analysts and Python data scientists.

Cloud Database ML Comparison

Capability	Google CloudBigQuery ML	AWSAmazon Redshift ML	SnowflakeCortex AI & Snowpark	AzureSynapse ML / Databricks
In-Database Engine	Native BQML SQL engine	Redshift ML (delegates to SageMaker)	Snowflake Cortex & Snowpark ML	SynapseML (Apache Spark based)
Workflow Approach	Pure SQL in warehouse	SQL calls SageMaker Autopilot	SQL & Snowpark Python	SQL, PySpark, Scala
GenAI & LLM Integration	Native Gemini & Vertex AI via SQL	Bedrock integration via SQL	Cortex LLMs (Llama, Mistral)	Azure OpenAI / Cognitive Services
Built-in Algorithms	Regression, Trees, ARIMA+, Matrix Fact.	AutoML via SageMaker Autopilot	Classification, Anomaly Detection	MLlib / SparkML library support
Setup Complexity	Zero infrastructure setup	Requires SageMaker & S3 permissions	Requires Cortex credits / setup	Requires Spark cluster provisioning

AWSEquivalent

Redshift ML runs CREATE MODEL in SQL, but exports data to SageMaker Autopilot in the background.

Snowflake Equivalent

Snowflake Cortex AI offers SQL functions for LLMs, while Snowpark ML serves Python developers.

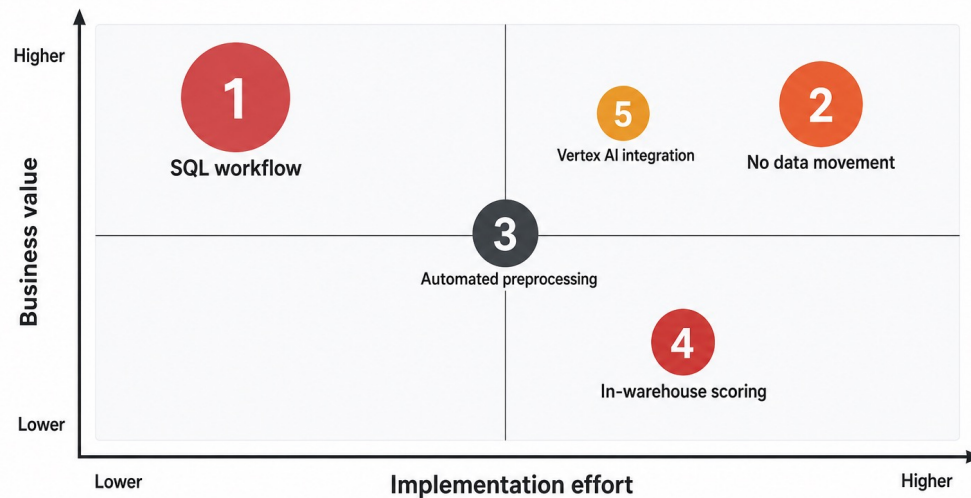
Key BQML Advantage

Most mature, fully integrated end-to-end SQL experience with zero cluster or resource provisioning required.

Key Features: Why BigQuery ML is a Game Changer

BigQuery ML Feature Value Matrix

Why data analysts can build and use ML without leaving the data warehouse



Quick wins: **1-3** | Production scale: **4-5**

Core Advantage

BigQuery ML democratizes machine learning by allowing data analysts to build models using standard SQL without moving data.

1. SQL-Based Workflow

Use familiar 'CREATE MODEL' statements to build sophisticated models.

2. No Data Movement

Eliminates security risks and latency of moving data to external platforms.

3. Automated Preprocessing

Built-in functions for feature engineering and handling missing values.

4. Model Deployment

Models are instantly available for real-time predictions as BQ functions.

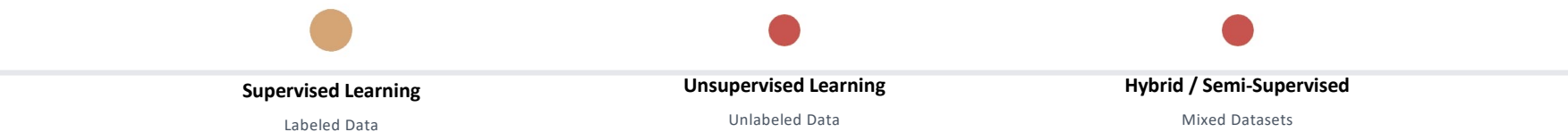
5. Vertex AI Integration

Seamlessly connects with Vertex AI for advanced model management.

Workflow Phases:

- Data Preparation
- Model Training
- Model Evaluation
- Real-time Prediction

The Machine Learning Landscape: Supervised vs. Unsupervised



Supervised Learning

Labeled Data

Unsupervised Learning

Unlabeled Data

Hybrid / Semi-Supervised

Mixed Datasets

Core Distinction

Supervised learning uses labeled datasets to train algorithms to classify data or predict outcomes, while unsupervised learning discovers hidden patterns in unlabeled data.

Key Learning Themes

- **Supervised:** Predictive modeling with known ground truth.
- **Unsupervised:** Exploratory analysis for structure discovery.
- **Hybrid:** Combining both for complex real-world insights.

Learning Method Highlights

1

Supervised: Regression

Predicting continuous values (e.g., price, temperature)

2

Supervised: Classification

Categorizing data into discrete classes (e.g., spam vs. not spam)

3

Unsupervised: Clustering

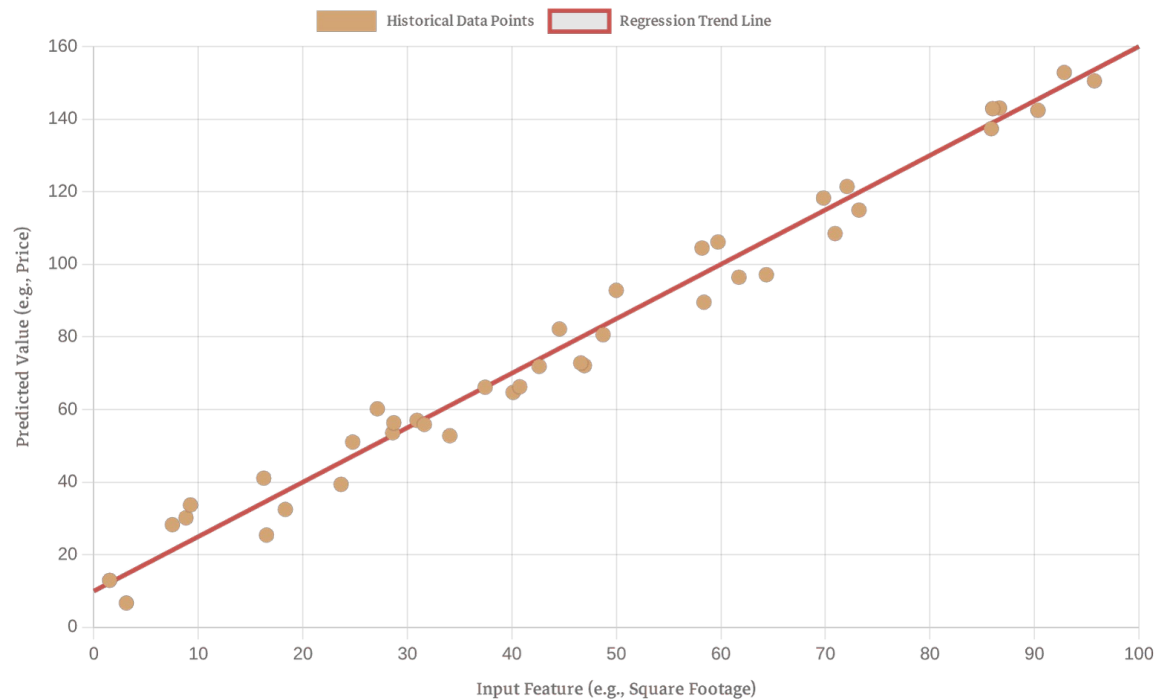
Grouping similar data points together (e.g., customer segments)

4

Unsupervised: Association

Discovering rules that describe data (e.g., market basket analysis)

Regression: Predicting Continuous Numerical Values



Source: BigQuery ML Documentation - Sample Regression Visualization

Key Insight

Regression models in BQML enable precise forecasting of numerical outcomes by analyzing historical relationships between variables.

Linear Regression

Best for simple, direct relationships between inputs and outputs.

Boosted Tree Regression

Handles complex, non-linear patterns with high accuracy.

ML.EVALUATE Metrics

Assess performance using R-Squared, Mean Absolute Error, and RMSE.

Common Use Cases:

- Real estate price estimation
- Sales and revenue forecasting
- Demand planning for inventory
- Financial risk assessment

Classification: Categorizing Data into Distinct Groups

Key Insight

Classification is a supervised learning method that assigns data points to predefined categories, enabling automated decision-making at scale.

Core Applications

- **Fraud Detection:** Identifying suspicious financial transactions.
- **Spam Filtering:** Categorizing emails as "Spam" or "Ham".
- **Customer Churn:** Predicting which users are likely to leave.
- **Sentiment Analysis:** Classifying text as positive or negative.

Boosted Trees

XGBoost integration in BQML

Binary Classification

Two-class (Yes/No) outcomes

Logistic Regression

The BQML Baseline

Deep Neural Networks

Complex non-linear patterns

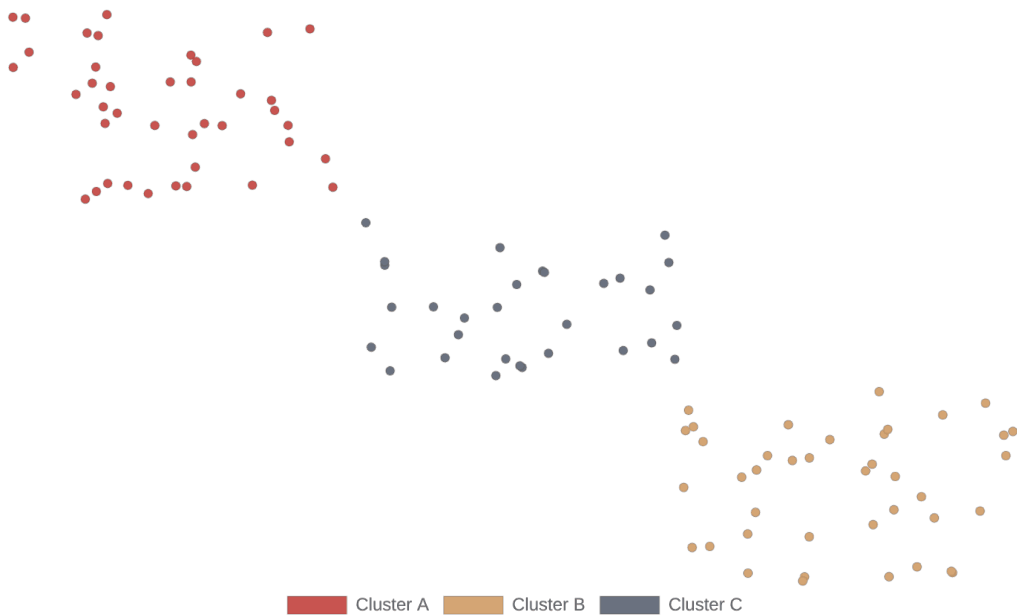
Multi-class

Predicting multiple categories

Random Forests

Ensemble learning for accuracy

Clustering: Discovering Natural Groups in Unlabeled Data



Visualizing K-Means Clustering: Grouping data points by similarity

Key Insight

Unsupervised learning in BigQuery ML reveals hidden structures in data without requiring predefined labels or explicit guidance.

K-Means

Primary BQML Algorithm

Anomalies

Fraud & Log Detection

Practical Use Cases:

- Customer Profiling
- Inventory Grouping

Segmentation

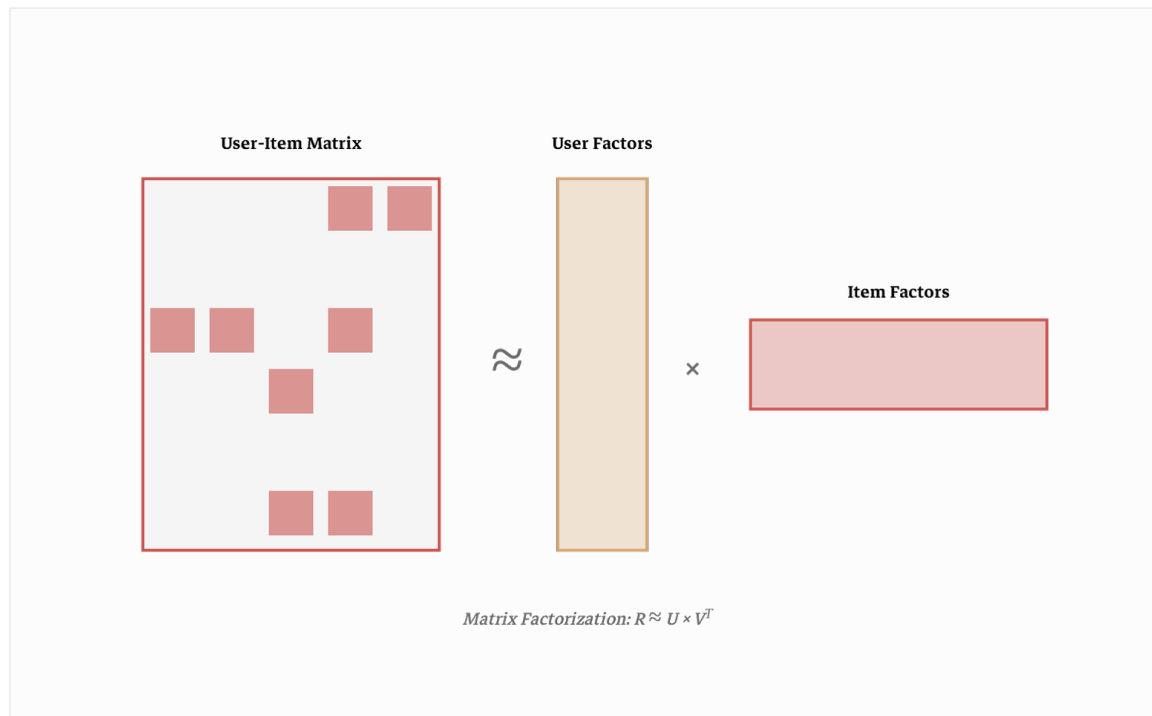
Targeted Marketing

Discovery

Behavioral Patterns

- Document Topic Discovery
- Network Security

Recommender Systems: Personalized User Experiences



Core Mechanism

BigQuery ML utilizes **Matrix Factorization** to decompose user-item interactions into latent factors, enabling highly personalized predictions.

Latent Factors discovery

Uncovering hidden preferences like genre or style

Collaborative Filtering

Leveraging "wisdom of the crowd" for accuracy

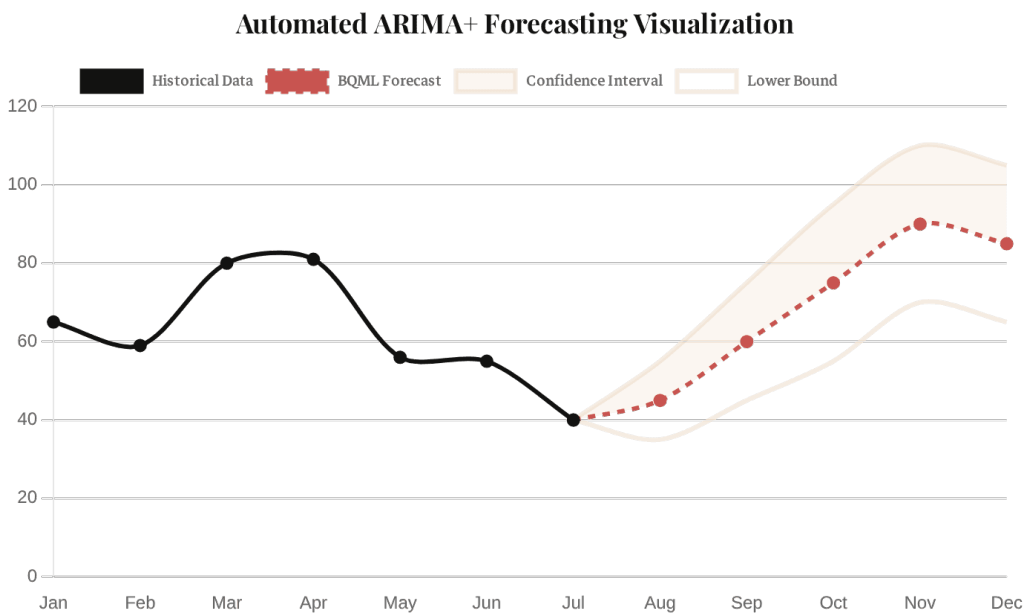
Scalable Predictions

Handling millions of users and items in SQL

Key Use Cases:

- E-commerce product suggestions
- Content & Media personalization
- Next-best-offer in banking
- Dynamic pricing & bundling

Time Series Analysis: Forecasting Future Trends



Source: BigQuery ML ARIMA+ Documentation & Sample Data

ARIMA+ Capabilities

BigQuery ML's automated time series forecasting handles seasonality, holidays, and outliers without manual tuning.

1,000+

Parallel time series

80-95%

Confidence intervals

50+

Country holidays

1 SQL

Command to forecast

Key Use Cases:

- Retail sales demand
- Energy consumption
- Inventory planning
- Network traffic

Workflow Complete

From Raw Data to Intelligent Predictions in SQL

The 5-Step BQML Workflow



1. Data Preparation

Clean and explore data using standard SQL.



2. Model Creation

Use `CREATE MODEL` with chosen algorithms.



3. Evaluation & Prediction

Assess with `ML.EVALUATE` and predict with `ML.PREDICT`.

Learning Resources

- [Official BigQuery ML Documentation](#)
- [Google Cloud ML Learning Path](#)
- [Vertex AI Integration Guide](#)
- [BQML SQL Reference & Tutorials](#)

"BigQuery ML democratizes machine learning by bringing the models to the data, not the data to the models."